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A phonetically based hypothesis on the evolution of *zi* rime change: Evidence from the Xiaxian dialect

基于语音的子变韵演化路径假说：来自夏县方言的证据

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Poster & References

Highlights

1. **Acoustic data confirm vowel lengthening in Xiaxian *zi* rime change in all four speakers.**
   声学数据证实：夏县子变韵中的元音延长见于全部四位发音人。
2. **Some speakers add a backed offglide (a short "tail") after high front vowels.**
   部分发音人在高、前元音之后出现一个后移滑音。
3. **This tail may be where the /u/ ending of other dialects comes from.**
   这个后移滑音可能是其他方言子变韵中 /u/ 尾的来源。

1. Background: *zi* rime change

Zi rime change (子变韵): the meaning of the Mandarin suffix *-zi* (子) is expressed by an alternation in the **rime of the root**.

Dialect	root	zi noun	type
Xunxian (Henan)	tʂuə^{24}	tʂuau^{24}	concatenation 拼合
Huojia (Henan)	tʂuaʔ^{33}	tʂuo^{33}	fusion 融合
Xiaxian (Shanxi)	pfʰɤ^{33}	pfʰɤː^{33}	lengthening 长音
Wenxi (Shanxi)	pfʰə^{53}	pfʰəːu^{51}	lengthening + concat. 长音+拼合

'table' in four dialects (example (1); F. Wang 1999 types)

Puzzle: In many dialects, the rime-changed forms end in a **back rounded** segment (/u/, /o/), yet *zi* suffixes contain an apical vowel or schwa ([tsɪ], [təʔ], [tə]). **Where does /u/ come from?**

Received pathway (after H. Wang 2004, 2008)

CV + tsɪ → … → CV + ə → CV:ə → **?** → CV:u → CVu (→ fusion) Henan; CV: Shanxi

- Length-only dialects such as **Xiaxian** = a **late stage**, after loss of /u/.
- The step ə → u (**?**) remains controversial.

2. Xiaxian zi rime change & RQs

Xiaxian (夏县), Yuncheng, southern Shanxi; Fenhe group of Zhongyuan Mandarin. Fieldwork in Yaofeng Town (瑶峰镇), 2024 & 2026.

- 36 basic finals; 32 have rime-changed forms
- Previous descriptions: lengthening only
- **New:** a backed offglide

	item	basic	rime-changed	gloss
a.	裤	kʰu^{33}	kʰuː^{33}	'pants'
	刀	tɑw^{52}	tɑːw^{52}	'knife'
	桌	pfʰɤ^{33}	pfʰɤː^{33}	'table'
b.	梯	tʰi^{52}	tʰiːX̱^{52}	'ladder'
	女	nᵚy^{23}	nᵚyːX̱^{23}	'girl'
	身	ʂẽj^{52}	ʂẽːX̱^{52}	'body'
	柜	kʰwej^{33}	kʰweːX̱^{33}	'wardrobe'

a. lengthening; b. lengthening + offglide.

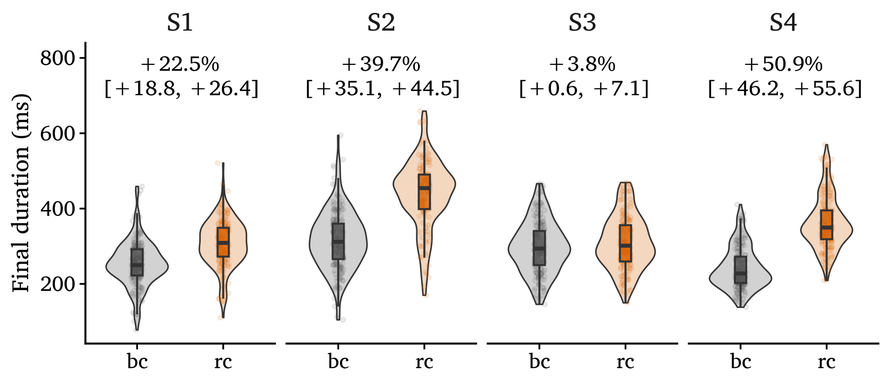
- [X̱] = **backed offglide**: auditorily central to back, with narrowing of lip aperture; salient for some speakers, weak or absent for others. Not reported in earlier descriptions.
- Its distribution appears regular, related to vowel height and frontness.

- **RQ1** Are rime-changed finals consistently longer than basic finals?
- **RQ2** What are the acoustic properties of the backed offglide, as reflected in formant trajectories?
- **RQ3** What are the phonetic conditions on its occurrence?

3. Method

- **Speakers:** S1–S4 (2 F, 2 M), aged 69–74, lifelong residents of Yaofeng Town.
- **Materials:** 59 items, 31 basic finals.
  - Each item as **basic form (bc)**, read in 这是 ___ 'This is ___', and **rime-changed form (rc)**, naming a pictured object in the same frame.
  - 3 repetitions.
- **Data:** 1,371 tokens after quality control (707 bc, 664 rc). MFA alignment, manual correction in Praat.
- **Measures:** final duration; F1, F2 at 20 time-normalized points.
- **Analyses:**
  - RQ1 mixed-effects model of log duration.
  - RQ2 offset backing and Δ F2 decline.
  - RQ3 Spearman ρ per speaker, Holm-adjusted.

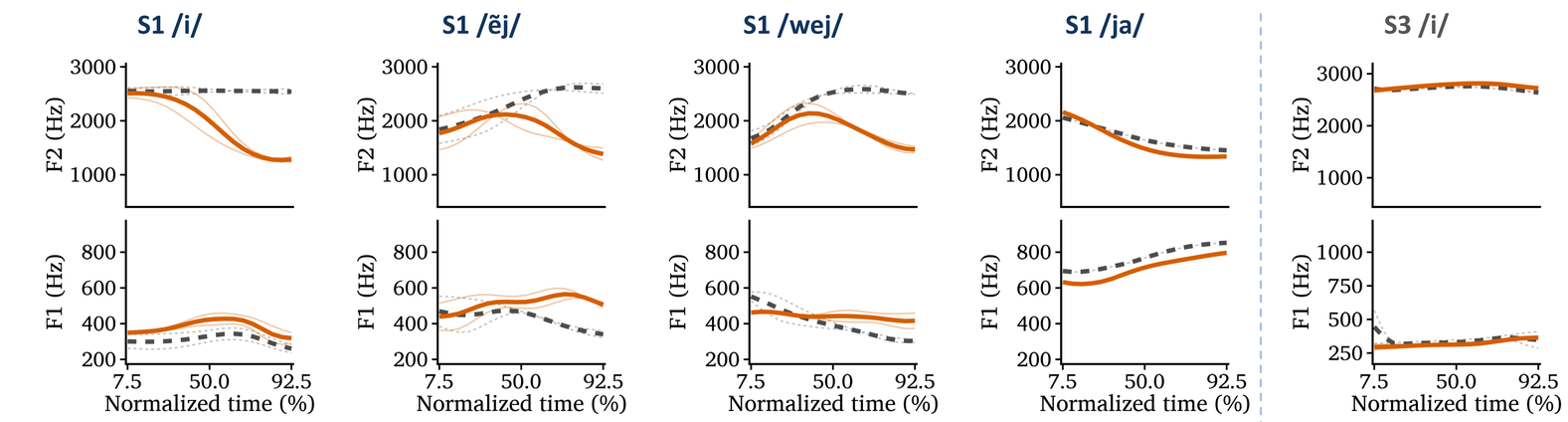
4. RQ1: Longer duration in *zi* rime change



- **All four speakers lengthen rime-changed finals**: +28.0% overall [25.8, 30.2].
- Magnitude varies by speaker: S3 +3.8%, S1 +22.5%, S2 +39.7%, S4 +50.9% (form × speaker $\chi^2(3) = 291.6$, $p < .001$); no tone effect.

bc = basic form; **rc = rime-changed form**; % = model-estimated lengthening [95% CI] (paper Fig. 1).

5. RQ2: The backed offglide in formant trajectories



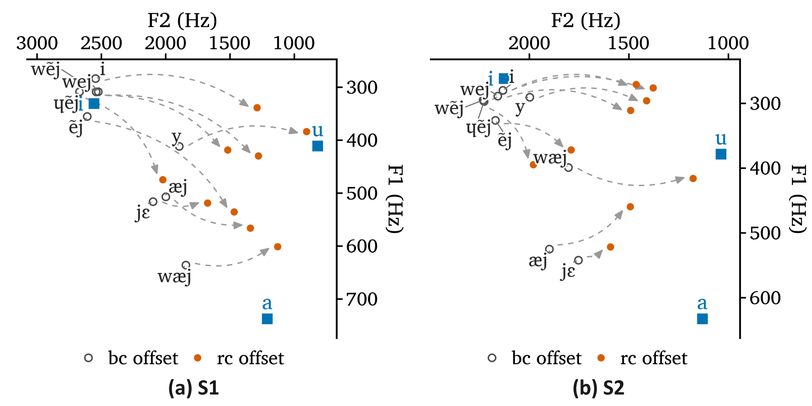
Mean F2 (upper) and F1 (lower) over normalized time. **Dashed grey = basic**, **solid orange = rime-changed**; thin lines = item means.

- **S1 and S2: clear F2 lowering toward the offset in some rime-changed finals.**
- **S3:** bc and rc trajectories largely overlap. **S4:** a weaker offglide, with smaller offset backing (e.g., /i/, /wẽj/).
- The offglide is **gradient and unstable**: its size varies by final and speaker, and end points do not converge on one target (they depend on the basic final). F1 changes too (S1, S4).

6. RQ3: The offglide occurs mainly after high front finals

Final	S1		S2		S3		S4	
	OB	ΔD	OB	ΔD	OB	ΔD	OB	ΔD
wẽj	1261	710	855	532	−26	−31	375	−40
i	1256	1180	669	490	−78	−23	278	147
ẽj	1143	671	380	271	−29	46	169	34
wej	1006	559	667	431	−70	−18	118	−34
y	990	430	587	304	695	58	201	12
wæj	713	97	626	243	−33	87	284	36
æj	658	569	406	112	−55	30	268	−21
ɥẽj	648	158	248	161	−124	2	−22	107
jɛ	424	295	161	63	97	176	191	265
u	194	358	135	166	−194	87	−30	55
a	24	−69	15	43	75	−30	−12	13
ej	−80	18	48	3	−84	−25	−64	17

OB = offset backing (bc − rc offset F2); **ΔD = Δ F2 decline** (rc − bc peak-to-offset F2 decline); Hz. Shading: ≥900, ≥600, ≥300, ≥100. Selected finals.



Basic and rime-changed offsets in F1–F2 space for the nine finals, S1 (a) and S2 (b); /i, u, a/ as reference.

- In **S1 and S2**, **/i, y, æj, wæj, ẽj, jɛ, wej, wẽj, ɥẽj/** combine offset backing with a larger F2 decline; rc offsets move toward the **central-to-back** region. **S4**: weak; **S3**: none (except one /y/ item).
- Larger changes after **higher, fronter** basic-form offsets (S1, S2; Spearman ρ = .60–.68, Holm $p < .05$).

7. Interpretation and an evolutionary hypothesis

Representation. The Xiaxian *zi* affix = a mora with a floating [+back]: **{μ, [+back]}** (see also Lin 1993; Yang 2024)

tʰi (basic form 'ladder') + {μ, [+back]} (μ → length; [+back] → direction) → tʰiːX̱ (rime-changed form)

- [+back] fixes the **direction**: displacement is larger after high front finals and vacuous after /a/.
- Speaker variation = how μ and [+back] are phonetically realized.

Hypothesized pathway

CV + tsɪ → … → CV + əᵘ — suffix reduced to schwa (+ mora)

→ **CV + {μ, [+back]} → CV:X̱** — gradient backed offglide (Xiaxian)

→ **CV:u** — offglide **reanalyzed** as a segment; [+round] enhances backness

→ **/u/ spreads** — by **analogy** to finals without an offglide

→ **CVu → fusion** — long vowel lost; /u/ may fuse (e.g., Huojia)

- **Consequence:** Xiaxian would sit **before** the /u/ ending arose, not after its loss, and the gap ə → u in the received pathway would be filled by listener reanalysis (Ohala 1993; Beddor 2009).

CV + tsɪ → … → CV + ə → CV:ə → **reanalysis analogy** → CV:u → CVu (→ fusion) *Henan*; CV: *Shanxi*

8. Closing remarks

- Xiaxian *zi* rime change = **lengthening** (all speakers) + a **backed offglide** (mainly S1, S2; only after high front basic finals).
- The offglide is **gradient, unstable** and **tied to morphology** → *zi* affix {μ, [+back]}.
- **Hypothesis:** the /u/ ending of other dialects may come from listeners' reanalysis of such a backing tendency, not from a suffix containing /u/.
- **Next:** end points vs. each speaker's /u/ and /ə/; lip-rounding data; other Shanxi lengthening dialects.